

Zhihao Zhu

Postdoctoral Fellow at HKUST

✉ zhihaozhu@ust.hk  [GitHub](#)  [Google Scholar](#)  [ORCID](#)

Experience

- **Postdoctoral Fellow**, Hong Kong University of Science and Technology, advised by Prof. Yi Yang.
- **Ph.D. in Computer Science**, University of Science and Technology of China (2025), advised by Prof. Defu Lian.
- **Bachelor's degree in Computer Science**, Fuzhou University (2020).

Research Focus

I study trustworthy AI and large language models, focusing on auditing, privacy, and training data exposure.

Trustworthy AI

Auditing model behavior, data revocation, membership inference, and model stealing risks in recommender and graph learning systems.

Large Language Models

Detecting training data exposure and understanding privacy risks in language and vision-language models, emphasizing pre-training data identification and leakage measurement.

Business & Interdisciplinary Journals

Forget Me If You Can: Auditing User Data Revocation in Recommendation Systems

Zhihao Zhu, Yi Yang, Yangyang Fan, Defu Lian

[Information Systems Research](#) 2026

HoneyImage: Verifiable, Harmless, and Stealthy Dataset Ownership Verification for Image Models

Zhihao Zhu, Jiale Han, Yi Yang

[Management Information Systems Quarterly](#) Major Revision

Revealing Training Data Exposure in Vision-Language Large Models via Parameter Gradients

Zhihao Zhu, Hongyi Tang, Yi Yang, Ahmed Abbasi

[Nature Communications](#) Major Revision

RecShield: Output-Level Attribute Unlearning in Recommender Systems

Zhihao Zhu, Yi Yang

[Information Systems Research](#) Under Review

GraphMSA: Stress Testing Graph Classification Services Against Model Stealing Attacks

Zhihao Zhu, Yi Yang, Chenwang Wu, Defu Lian

[INFORMS Journal on Computing](#) Under Review

AI Conferences and Journals

TDDBench: A Benchmark for Training Data Detection

Zhihao Zhu, Yi Yang, Defu Lian

[International Conference on Learning Representations](#) 2025

Identifying Pre-training Data in LLMs: A Neuron Activation-Based Detection Framework

Hongyi Tang*, Zhihao Zhu*, Yi Yang. Equal contribution.

[Conference on Empirical Methods in Natural Language Processing](#) 2025

Membership Inference Attacks against Sequential Recommender Systems

Zhihao Zhu, Chenwang Wu, Rui Fan, Defu Lian, Enhong Chen

[The Web Conference](#) 2023

Resisting Graph Adversarial Attack via Cooperative Homophilous Augmentation

Zhihao Zhu, Chenwang Wu, Min Zhou, Hao Liao, Defu Lian, Enhong Chen

[European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases](#) 2022